Freight Rate Prediction

Model Development, Validation, and December 2025 Forecast

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Final selected approach: one CatBoost regression model trained on a common set of 16 features available in both the validation and December input datasets.

Chronological holdout performance: MAE = 122.58, RMSE = 633.90, $R^2 = 0.8275$.

Abstract

This project develops a supervised machine-learning model to estimate posted freight rates from shipment attributes. The labelled development dataset contains 48,000 loads dated from January through October 2025. A chronological holdout was used to approximate prediction on later observations: records before 1 September 2025 were used for model fitting and records dated 1 September through 31 October 2025 were used for validation. Data inspection identified missing and physically invalid negative shipment weights, a right-skewed target distribution, and a strong relationship between distance and posted rate. Feature engineering added squared distance, distance per unit weight, route, and calendar attributes. Several baselines and boosting models were compared. Although a full-feature engineered CatBoost model achieved the lowest validation MAE, the final operational choice was a common-feature CatBoost model because it uses only variables available in both the 12,000-row prediction file and the fixed December chart input. The selected model achieved MAE = 122.58, RMSE = 633.90, and $R^2 = 0.8275$ on the chronological holdout. It was then refitted on all 48,000 labelled records and used to generate the required validation and December predictions. The supplied scorer confirmed the structure and basic validity of both output files and generated the December chart.

Keywords: freight-rate prediction, regression, CatBoost, gradient boosting, chronological validation, feature engineering.

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1 Problem Statement and Deliverables

The objective is to estimate the posted rate for each freight load using historical labelled shipment data. The development file, `data/train_test.csv`, contains 48,000 labelled records. The prediction file, `data/validation.csv`, contains 12,000 unlabelled loads with unique load identifiers. A separate 31-row December input specifies one fixed lane and shipment configuration for each day from 1 December to 31 December 2025.

The implemented workflow addresses three outputs:

1. A CSV named `validation_predictions.csv` with exactly the columns `load_id,predicted_rate` and 12,000 rows.
2. A completed `data/december_chart_inputs.csv` containing the original six input fields and the additional `predicted_rate` column.
3. The scorer-generated chart `scorer_results/candidate_december.png`.

The assessment also requests a reproducible code repository, dependencies and run instructions, a report describing the split and validation approach, and a short walkthrough video. This report documents the modelling workflow and the observed results. The scorer's messages confirm file validation and chart generation; they do not provide a final external accuracy score.

2 Dataset Overview

2.1 Development data

The labelled development data has 48,000 rows and 14 columns. The columns are `load_id`, `pickup`, `delivery`, four geographic coordinate fields, `distance`, `equipment`, `weight`, `date`, `market_index`, `quote_signal`, and the target `posted_rate`. The development dates run from 1 January 2025 through 31 October 2025.

The prediction dataset contains 12,000 rows for which the model must generate a rate. The December chart input contains 31 daily rows and provides only `pickup`, `delivery`, `distance`, `equipment`, `weight`, and `date` before the prediction column is filled.

2.2 Target distribution

The observed target summary is shown in Table 1. The mean is above the median and the maximum is far above the upper quartile, indicating a right-skewed distribution with a long upper tail.

Table 1: Summary statistics for `posted_rate` in the 48,000-row development dataset.

Statistic	Value
Count	48,000
Mean	2,373.98
Standard deviation	1,486.49
Minimum	57.22
25th percentile (Q1)	1,251.56
Median	2,030.76
75th percentile (Q3)	3,330.75
Maximum	25,533.00
Interquartile range (IQR)	2,079.20

The high-rate tail was not removed solely because it appeared as statistical outliers. Such records may represent genuine freight transactions, and deleting them without evidence could distort the

prediction task. The later error analysis shows that rare, high-rate validation records are difficult for the model and should be treated as an important limitation.

2.3 Distance, equipment, and market attributes

Distance has a mean of 1,135.86, standard deviation of 728.56, median of 953.30, minimum of 70.00, and maximum of 3,439.80. The equipment-level rate summaries are shown in Table 2.

Table 2: Posted-rate summaries by equipment type.

Equipment	Loads	Mean rate	Median rate	Std. dev.
Dry Van	27,202	2,271.55	1,953.04	1,423.03
Flatbed	8,753	2,445.09	2,076.81	1,505.05
Reefer	12,045	2,553.64	2,196.67	1,589.67

Other numeric summaries from the development data are: weight mean 31,028.84, standard deviation 9,391.44, median 31,436.50, minimum $-47,500$, and maximum 47,500; market index mean 1.0834, standard deviation 0.1681, minimum 0.6764, maximum 1.4678; and quote signal mean 2.0625, standard deviation 0.2914, minimum 0.6923, maximum 3.6104.

2.4 Exploratory correlations

The Pearson correlation with `posted_rate` was approximately 0.909 for distance, making it the strongest observed linear association. Delivery longitude and pickup longitude had correlations of approximately -0.257 and -0.255, respectively. Delivery latitude and pickup latitude were approximately -0.092 and -0.091. Weight and market index had weak positive correlations of approximately 0.035 and 0.034, while quote signal had a weak negative correlation of approximately -0.040. These are univariate linear correlations and do not prove that features with weak correlations are unhelpful to a nonlinear model.

The monthly mean and median rates moved in broadly similar directions during the observed period: rising into June, declining through August, recovering in September, and easing slightly in October. This pattern may be affected by changes in the composition of routes and shipment types; it should not be interpreted as an isolated market-seasonality effect.

2.5 Exploratory price-per-distance ratio

The exploratory ratio `posted_rate/distance` had the following mean and median values by equipment: Dry Van 2.115 and 2.046; Flatbed 2.295 and 2.220; Reefer 2.383 and 2.312. This ratio was used only for exploratory analysis. Because it is calculated using the target `posted_rate`, it was not supplied as a model input; including it would cause target leakage.

3 Data Quality and Preprocessing

3.1 Missing and invalid weights

The development data contained 300 missing weights (0.625%) and 292 negative weights (0.608%). Negative shipment weights are physically implausible. The negative values were not converted to positive values; instead, they were treated as missing. Two indicators were created before replacement: `weight_was_negative` and `weight_was_missing`. After invalid negative values were set to missing, the cleaned training weight column contained 592 missing values. In the 12,000-row prediction data, 165 weights were missing and 145 were negative.

The target distributions for missing, negative, and valid-weight rows were similar in the initial inspection: the missing-weight group had mean rate 2,350.67 and median 1,985.00; the negative-weight

group had mean 2,389.79 and median 1,994.96; and the valid-weight group had mean 2,374.03 and median 2,031.44. This supported retaining the records and using indicator features rather than dropping the rows.

Table 3: Weight-quality observations in the development data.

Condition	Rows	Share of 48,000
Missing weight in raw data	300	0.625%
Negative weight in raw data	292	0.608%
Valid weight	47,408	98.767%

Market index had 374 missing values (0.779%) in the development data. A `market_index_missing` indicator was created for experiments using the full feature set. The final common-feature model does not use market index because it is not provided in the December chart input.

3.2 Date and categorical preprocessing

The date field was parsed as a datetime. The following calendar features were extracted: `year`, `month`, `day`, `day_of_week`, `day_of_year`, and `is_weekend`. A route feature was constructed by joining pickup and delivery city names. Textual categorical values were represented as strings for CatBoost's categorical-feature interface.

3.3 Geographic coordinates

All four coordinate columns had 48,000 non-missing values in the development data. The observed pickup latitude range was 28.35765 to 44.30296 and pickup longitude range was -121.69849 to -69.50000. Delivery latitude and longitude had the same respective extrema. Range checks found no values outside the mathematical latitude/longitude bounds; such a check does not guarantee that each coordinate is geographically correct for its city. The coordinate fields were excluded from the final common-feature model because they are absent from the December input file.

4 Feature Engineering and Model Inputs

Two additional numerical features were tested:

$$d_{\text{sq}} = d^2, \tag{1}$$

$$r_{dw} = \frac{d}{w}. \tag{2}$$

Here, d is distance and w is shipment weight; the resulting columns are named `distance_squared` and `distance_per_weight`. Zero weights were treated as missing for the ratio calculation. Missing values in the ratio were left for CatBoost's missing-value handling. The raw target-derived ratio `price_per_distance` was explicitly excluded.

The final operational model uses 16 features that can be produced for both prediction tasks:

- Shipment and lane: `pickup`, `delivery`, `distance`, `equipment`, `weight`, and `route`.
- Weight-quality indicators: `weight_was_negative` and `weight_was_missing`.
- Calendar attributes: `year`, `month`, `day`, `day_of_week`, `day_of_year`, and `is_weekend`.
- Engineered numeric features: `distance_squared` and `distance_per_weight`.

Pickup/delivery coordinates, market index, quote signal, the market-index missingness flag, the identifier, the raw date, and the target itself were not model features in the final common-feature configuration. The raw date was represented through derived calendar attributes. The identifier `load_id` was retained only to map predictions back to rows.

5 Validation Design

A chronological split was selected because the final task predicts later loads from historical records. The development data was sorted by date. All rows dated before 1 September 2025 formed the training partition; rows dated on or after 1 September 2025 formed the holdout partition. Since the development data ends on 31 October 2025, the holdout covers September and October.

Table 4: Chronological development-data split.

Partition	Date range	Rows
Training	1 Jan–31 Aug 2025	38,477
Validation	1 Sep–31 Oct 2025	9,523
Total	1 Jan–31 Oct 2025	48,000

The feature matrix and target vector were rebuilt after chronological sorting, and their indices were checked for alignment. This prevents an ordering mismatch between the feature rows, target values, and date-based split. The 9,523-row holdout was used for model comparison. After choosing the common-feature model, the model was refitted on all 48,000 labelled records for inference on the supplied prediction files.

6 Models and Validation Results

6.1 Evaluation metrics

The experiments were compared using mean absolute error (MAE), root mean squared error (RMSE), and coefficient of determination (R^2):

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|,$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}.$$

MAE measures average absolute prediction error. RMSE penalizes larger errors more strongly. R^2 measures the fraction of target variance explained relative to a mean-only reference; it can be negative when a model performs worse than that reference.

6.2 Experiment comparison

Table 5 summarizes the reported chronological-holdout results. All values are the outputs recorded during the model experiments; small differences across separate runs should be interpreted in light of the specific feature configuration used in each run.

Table 5: Model comparison on the 9,523-row chronological validation holdout. Lower MAE and RMSE are better; higher R^2 is better.

Model / experiment	MAE	RMSE	R^2
Mean baseline	1,178.8033	1,526.2263	-0.0002
Original CatBoost, full features	113.9046	633.6758	0.8276
Log-target CatBoost, full features	113.5560	635.4788	0.8266
XGBoost	163.9005	663.6766	0.8109
Engineered CatBoost, full features	**109.2419**	**633.0680**	**0.8279**
Common-feature CatBoost	122.5832	633.8997	0.8275
December-compatible CatBoost run	122.9862	633.4106	0.8277

The engineered full-feature CatBoost run had the lowest validation MAE and RMSE among the reported experiments. However, that full feature set depends on fields such as coordinates, market index, and quote signal that are unavailable in the December chart input. The common-feature CatBoost model was therefore selected as the single operational model to keep the feature interface consistent for both required prediction tasks. Its MAE was about 12.2% higher than the engineered full-feature experiment, while its RMSE and R^2 remained close to the full-feature results.

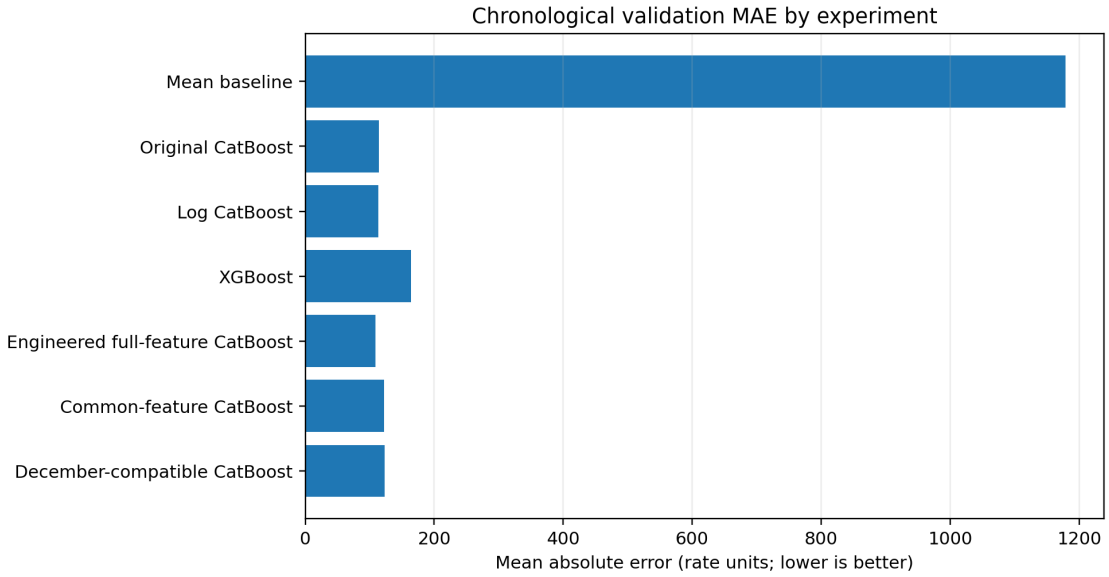


Figure 1: MAE comparison across the reported modelling experiments. This chart visualizes the values in Table 5.

6.3 Why CatBoost

CatBoost was chosen for the final common-feature model because the input contains categorical variables such as pickup, delivery, equipment, and route, in addition to numeric shipment and calendar attributes. It supports categorical features directly and produced materially lower errors than the mean baseline and the tested XGBoost configuration. The log-target experiment produced only a marginally lower MAE than the original CatBoost run while having worse RMSE and R^2, so the log transformation was not retained.

7 Feature Importance

The first full-feature CatBoost model assigned approximately 87.92 importance points to distance, followed by equipment (4.20), weight (1.32), market index (1.16), day of year (0.86), month (0.81), and quote signal (0.55). This indicated that distance dominated that model's split-based importance measure.

After adding the engineered features, the full-feature engineered CatBoost model assigned 49.68 importance points to squared distance and 31.86 to raw distance. Equipment contributed 5.23, weight 2.06, market index 1.86, and distance per weight 1.53. The two distance-derived features together accounted for approximately 81.54 importance points.

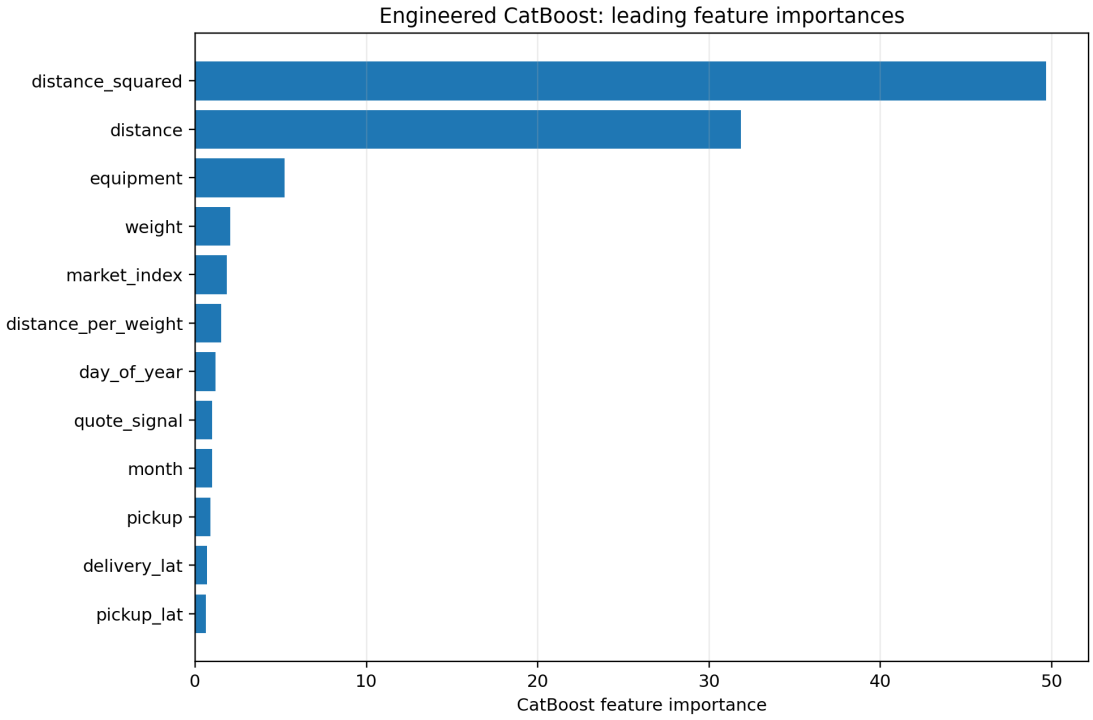


Figure 2: Leading feature importances from the engineered full-feature CatBoost model. The chart uses the feature-importance values reported by the notebook.

Feature importance is model-dependent and describes the model's allocation of predictive importance; it does not establish causality. Low importance in one fitted model does not prove that a feature is universally useless. The common-feature set was selected primarily to make the same model usable with the December input schema, not simply by deleting every low-importance feature.

8 Error Analysis and Limitations

An error-band analysis of the initial full-feature CatBoost validation run indicated that most validation observations were below 7,000 rate units, where the average absolute errors were comparatively small. The upper tail contained only 41 observations across the two bands above 7,000, and errors were substantially larger.

Table 6: Error analysis by actual-rate band for the initial full-feature CatBoost validation run.

Actual-rate band	Count	MAE	Mean actual	Mean predicted
0–1,000	1,545	81.08	697.32	758.22
1,000–2,000	3,077	47.58	1,482.62	1,480.84
2,000–4,000	3,387	69.91	2,845.52	2,836.96
4,000–7,000	1,473	160.51	4,770.81	4,684.72
7,000–10,000	17	5,508.66	8,392.65	2,883.99
Above 10,000	24	10,256.90	14,003.43	3,746.53

The high-rate bands are small and the corresponding mean predictions are much lower than the mean actual rates, indicating severe underprediction in this rare tail. This is consistent with the substantial gap between MAE and RMSE. This grouped diagnostic belongs to the initial full-feature CatBoost run and should not be interpreted as a separately measured error breakdown for the final common-feature model.

Other limitations include:

- The final common-feature model does not use coordinates, market index, or quote signal because those fields are absent from the December input. This keeps inference consistent but sacrifices information available for the full-feature experiment.
- The December chart represents one fixed lane and shipment configuration. Its daily pattern should not be generalized to all lanes or interpreted as verified market movement.
- Calendar features can create structured day-to-day fluctuations. Those patterns are model predictions, not observed December transaction prices.
- The supplied scorer validates file structure, required identifiers, positive numeric predictions, the 31 required dates, fixed shipment attributes, and chart generation. It explicitly states that final validation metrics are calculated by the assessment platform after submission. Those external metrics were not available during this work.
- Coordinate range checks only establish that values fall within valid numeric bounds; they do not confirm that every city-coordinate pair is geographically accurate.

9 Final Model and Prediction Outputs

The selected common-feature CatBoost model was refitted on all 48,000 labelled development records. The reported final estimator configuration was:

- Algorithm: CatBoost regressor.
- Iterations: 175 (the selected iteration count was the best validation iteration plus one).
- Learning rate: 0.05.
- Tree depth: 8.
- Training loss: RMSE.
- Random seed: 42.
- Input features: 16 common features listed in Section 5.

The 12,000-row prediction file was saved with exactly two columns, `load_id` and `predicted_rate`. The December file retained exactly seven columns in the required order: `pickup`, `delivery`, `distance`, `equipment`, `weight`, `date`, and `predicted_rate`.

The December scorer inputs were fixed at Lexington to Fort Wayne, distance 360 miles, Dry Van equipment, and weight 32,000 lb, with one row for each date from 1 December through 31 December 2025. The generated chart is shown in Figure 3.

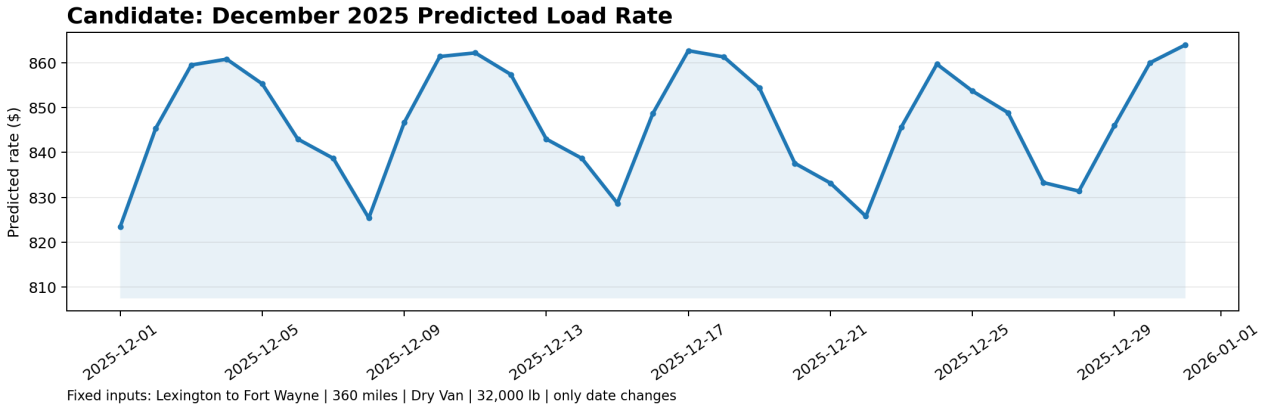


Figure 3: December 2025 predicted load-rate chart for the fixed Lexington–Fort Wayne lane. The included image is a visual reconstruction from the chart screenshot shared during the project discussion; replace it with the exact `scorer_results/candidate_december.png` file before final submission if exact pixel-for-pixel reproduction is required.

The displayed chart shows recurring rises and falls through the month, with plotted values visually around the low-to-mid $800s. Since the underlying 31 numeric predictions were not supplied separately for this report-generation step, no exact chart minimum, maximum, or daily values are asserted here. The exact chart created by the scorer should be used as the final report figure.

9.1 Scorer validation

The scorer completed successfully and printed:

> Validated 12,000 final predictions.
> Validated 31 fixed December predictions.
> Created chart: scorer_results
> candidate_december.png
> Final validation metrics are calculated by Spotter after submission.

This confirms structural and value checks passed for the two output files. It is not an independent confirmation of predictive accuracy.

10 Reproducibility

The executed workflow used Python with pandas, NumPy, Matplotlib, scikit-learn, CatBoost, and XGBoost for the model experiments. Exact package versions were not recorded in the available run log and are therefore not stated here. The project should include a pinned dependency file generated from the working environment before publication.

The scorer command used was:

```
python score.py --predictions validation_predictions.csv \
  --december-predictions data/december_chart_inputs.csv
```

The report's final output paths are:

- `validation_predictions.csv`
- `data/december_chart_inputs.csv`
- `scorer_results/candidate_december.png`

For a fully reproducible repository, include the notebook or scripts that perform preprocessing, chronological splitting, feature construction, model fitting, prediction generation, and the scorer invocation. The final CSVs should be generated by those scripts rather than manually edited.

11 Conclusion

A chronological validation strategy was used to reflect the time-ordered nature of the prediction task. Exploratory analysis found a right-skewed posted-rate target, missing and negative shipment weights, equipment-level rate differences, and a strong relationship between distance and posted rate. Feature engineering, especially squared distance, improved the full-feature CatBoost experiment. The engineered full-feature model recorded the best validation MAE, but it depended on variables unavailable in the December chart input. A common 16-feature CatBoost model was therefore selected to support both prediction tasks with one consistent feature interface. On the chronological holdout, this model achieved MAE = 122.58, RMSE = 633.90, and $R^2 = 0.8275$. The final estimator was refitted on all 48,000 labelled rows and generated both required prediction files. The provided scorer validated 12,000 validation predictions and 31 December rows and generated the chart. External post-submission accuracy metrics remain pending.

A Full Engineered-Model Feature Importance

Table 7 lists all values reported for the engineered full-feature CatBoost model. The values are the model's reported importance scores and sum to approximately 100.

Table 7: Reported feature importance for the engineered full-feature CatBoost model.

Feature	Importance
distance_squared	49.682440
distance	31.857045
equipment	5.232458
weight	2.060891
market_index	1.856417
distance_per_weight	1.525363
day_of_year	1.195667
quote_signal	1.004105
month	0.978321
pickup	0.885218
delivery_lat	0.689469
pickup_lat	0.624990
delivery	0.590078
day	0.478885
route	0.416766
delivery_lon	0.291099
pickup_lon	0.223777
is_weekend	0.204663
day_of_week	0.137940
weight_was_negative	0.060266
market_index_missing	0.004141
weight_was_missing	0.000000

B Final Common-Feature List

1. pickup
2. delivery
3. distance
4. equipment
5. weight
6. weight_was_negative
7. weight_was_missing
8. route
9. year
10. month
11. day

12. `day_of_week`
13. `day_of_year`
14. `is_weekend`
15. `distance_squared`
16. `distance_per_weight`